

Unit 1 – Introduction to Cyber Crime

Detailed 10–15 Marks Exam Notes

The topics in your syllabus are:

Definition and Origins of Cyber Crime

Introduction to Cyber Crime

Classification of Cyber Crime

Against Individual

Against Property

Against Government

Against Society

Global Perspective on Cyber Crime

Cyber Crime Transcends National Boundaries

Case Studies of Cyber Crime

Global Struggle with Cyber Security

Case Study related to Global Cyber Security

International Cooperation to Combat Cyber Crime

Case Study of International Cooperation

Planning of Cyber Offences by Criminals

1. Definition and Origins of Cyber Crime

Definition

Cyber crime refers to any illegal activity in which a computer, computer network, digital device, or the Internet is used as a tool, target, or medium for committing a crime.

Simple definition

Cyber crime = Crime committed using or against computers, networks, or digital systems.

Examples include:

Hacking

Phishing

Identity theft

Online fraud

Malware attacks

Cyber stalking

Ransomware

Data theft

Denial-of-Service attacks

Important point

A computer can play three different roles in cyber crime:

Role	Meaning	Example
Target	Computer/system itself is attacked	Hacking a server
Tool	Computer is used to commit crime	Online banking fraud
Storage	Computer stores illegal/stolen information	Storing stolen databases
Origins of Cyber Crime		

Cyber crime developed along with the development of computers and the Internet.

1. Early computer misuse

During the 1960s and 1970s, computers were mainly used by governments, universities and large organizations.

Unauthorized access to computer systems began appearing as computer networks became more common.

2. Phone/Phreaking era

Before widespread Internet usage, criminals experimented with telephone systems.

This activity became known as phreaking.

Attackers attempted to manipulate telephone networks to obtain unauthorized access or free services.

3. Computer viruses

During the 1980s, computer viruses became an important form of computer-related crime.

A virus could:

Replicate itself

Infect other files

Damage systems

Modify data

Disrupt computer operations

4. Internet expansion

During the 1990s, Internet usage expanded rapidly.

This created new opportunities for:

Email fraud

Hacking

Website attacks

Online scams

Identity theft

5. Modern cyber crime

With smartphones, cloud computing, social media, cryptocurrency and IoT, cyber crime has become more sophisticated.

Modern attacks include:

Ransomware

Advanced phishing

Cryptocurrency scams

Supply-chain attacks

Social engineering

State-sponsored cyber attacks

Deepfake-enabled fraud

Characteristics of Cyber Crime

Technology dependent

Can be committed from anywhere

Criminals can hide their identity

Can affect millions of people simultaneously

Evidence is mainly digital

Attackers can operate across international borders

Crimes can happen very quickly

Investigation requires specialized technical knowledge

Conclusion

Cyber crime originated with the misuse of early computer and communication systems and has evolved with technological development. Today, it is a global problem affecting individuals, businesses, governments and society.

Exam ending line

Thus, the growth of digital technology has created enormous benefits but has also created new opportunities for criminals, making cyber crime one of the major challenges of the digital age.

2. Introduction to Cyber Crime

Meaning

Cyber crime is a category of crime involving computers, networks, digital devices and the Internet.

Unlike traditional crimes, cyber crimes can often be conducted without the criminal being physically present at the location of the victim.

Example

A criminal sitting in one country can steal money from the bank account of a person in another country using the Internet.

Why Cyber Crime is Increasing

1. Rapid Internet growth

Millions of people use the Internet for:

Banking

Shopping

Education

Communication

Entertainment

This provides criminals with a large target population.

2. Digitalization

Organizations increasingly store information digitally.

Examples:

Customer information

Bank information

Medical records

Government records

Business secrets

Therefore, valuable information has become a target.

3. Easy availability of technology

Attackers can obtain sophisticated tools and information online.

4. Anonymity

Criminals can use techniques such as:

Fake accounts

Compromised systems

Proxy servers

Other anonymizing technologies

to make investigation more difficult.

5. Financial motivation

A large portion of cyber crime is financially motivated.

Examples:

Banking fraud

Ransomware

Credit-card fraud

Cryptocurrency scams

Common Types

Hacking

Unauthorized access to a computer or network.

Phishing

Fake emails/messages/websites are used to trick victims into revealing information.

Malware

Malicious software designed to damage, disrupt, spy on, or gain unauthorized access to systems.

Ransomware

Malware that makes data or systems inaccessible and demands payment.

Identity theft

Using another person's personal information without authorization.

Cyber stalking

Repeated unwanted monitoring or harassment through digital technologies.

Effects of Cyber Crime

Individual level

Financial loss

Privacy violation

Identity theft

Emotional distress

Business level

Financial loss

Data breaches

Reputation damage

Operational disruption

Government level

National security risks

Critical infrastructure disruption

Theft of sensitive information

Society level

Loss of trust in digital services

Economic damage

Social instability

3. Classification of Cyber Crime

Cyber crimes can broadly be classified according to the target/victim.

The four important categories in your syllabus are:

Cyber crime against individuals

Cyber crime against property

Cyber crime against government

Cyber crime against society

3.1 Cyber Crime Against Individuals

Definition

Cyber crimes against individuals are crimes where a specific person is the primary target.

Examples

Identity theft

Cyber stalking

Online harassment

Phishing

Email spoofing

Online cheating

Defamation

Account takeover

Cyber bullying

1. Identity Theft

Criminal obtains personal information such as:

Name

Password

Bank details

Government ID information

and uses it without authorization.

Example

An attacker obtains someone's login credentials and uses the account to make fraudulent transactions.

2. Cyber Stalking

Cyber stalking involves repeated unwanted monitoring, communication, harassment or intimidation through digital technologies.

It may occur through:

Social media

Email

Messaging platforms

Online forums

3. Cyber Bullying

Use of digital platforms to:

Threaten

Humiliate

Harass

Intimidate

another person.

4. Phishing

Attackers create fake messages or websites that appear legitimate.

Example:

"Your bank account will be blocked. Click here to verify."

The victim enters credentials into the fake website.

Effects

Financial loss

Loss of privacy

Psychological stress

Reputation damage

Unauthorized account access

3.2 Cyber Crime Against Property

Definition

These crimes target digital or financial property.

Property may include:

Money

Databases

Software

Intellectual property

Digital assets

Websites

Computer systems

Major Examples

1. Data Theft

Unauthorized copying or stealing of confidential information.

2. Intellectual Property Theft

Stealing:

Software

Source code

Designs

Research

Trade secrets

3. Ransomware

Attackers encrypt or otherwise deny access to data and demand payment.

4. Website Defacement

Unauthorized modification of a website.

For example:

Original website



Unauthorized access



Website modified



Attacker's message displayed

5. Computer Sabotage

Intentional damage or disruption of computer systems.

Effects

Financial losses

Loss of business data

Business interruption

Loss of intellectual property

Recovery costs

Reputation damage

3.3 Cyber Crime Against Government

Definition

Cyber crimes against government target:

Government websites

Government databases

Military systems

Critical infrastructure

Public services

Government communication systems

These attacks can become national security issues.

Major Examples

1. Cyber Espionage

Unauthorized access to systems to obtain sensitive information.

2. Cyber Terrorism

Use of cyber attacks with the objective of causing serious disruption or creating fear in connection with broader political or ideological aims.

3. Website Attacks

Government websites may be attacked or defaced.

4. Critical Infrastructure Attacks

Potential targets include:

Electricity

Transportation

Telecommunications

Water systems

Healthcare systems

Effects

National security risks

Economic losses

Disruption of public services

Theft of sensitive information

Damage to critical infrastructure

3.4 Cyber Crime Against Society

Definition

Cyber crimes against society affect large groups of people or society as a whole.

Examples

Online misinformation and malicious manipulation

Distribution of illegal content

Online financial scams

Cyber terrorism

Large-scale fraud

Organized cyber crime

Attacks on public services

Effects

Social disruption

Financial losses

Loss of public trust

Threats to public safety

Damage to social harmony

Increased fear and insecurity

Easy Classification to Remember

Remember:

I – P – G – S

I = Individual

P = Property

G = Government

S = Society

4. Global Perspective on Cyber Crime

Introduction

Cyber crime is no longer limited to one country or region.

The Internet connects computers across the world, allowing criminals to attack victims regardless of geographical boundaries.

Therefore:

Cyber crime is a global problem requiring global cooperation.

Why Cyber Crime is Global

1. Global Internet Connectivity

An attacker can operate remotely from another country.

2. International Criminal Networks

Cyber criminals can work together across different countries.

3. International Victims

One attack can affect users from multiple countries.

4. Cross-border infrastructure

Attackers may use systems located in several countries during a single operation.

For example:

Attacker



Compromised System A



Compromised System B



Victim Server

The attacker and victim may be in completely different countries.

Global Impact

Economic impact

Cyber crime causes:

Financial fraud

Business interruption

Data recovery costs

Cybersecurity expenditure

National security

Governments must protect:

Military systems

Government databases

Critical infrastructure

Privacy

Personal information can be stolen and misused internationally.

Challenges

Different laws in different countries

Difficulty identifying attackers

Jurisdiction problems

Different investigation procedures

Difficulty collecting evidence from another country

Rapid technological development

5. Cyber Crime Transcends National Boundaries

This is an important 10–15 mark question.

Meaning

A cyber crime is said to transcend national boundaries when different parts of the crime occur in different countries.

Unlike traditional crimes, physical distance is often not a major barrier.

Example

Suppose:

Criminal is in Country A

Server is in Country B

Victim is in Country C

Stolen money goes to Country D

Then:

Country A

Attacker



Country B

Server



Country C

Victim



Country D

Financial destination

One cyber crime can therefore involve four jurisdictions.

Why This Creates Problems

1. Jurisdiction

Which country's police should investigate?

2. Different laws

An action may be illegal in one country but treated differently in another.

3. Evidence collection

Digital evidence may be stored on servers in another country.

4. Attribution

Identifying the actual attacker can be difficult.

5. International cooperation

Authorities may need assistance from foreign agencies.

Solution

Countries need:

International treaties

Information sharing

Mutual legal assistance

Joint investigations

Cybersecurity cooperation

Common legal frameworks

Digital forensic cooperation

Conclusion

The borderless nature of cyberspace makes cyber crime a transnational problem, and effective investigation requires cooperation between countries.

6. Case Studies of Cyber Crime

Case studies are useful in exams because they demonstrate how cyber crime works in the real world.

Case Study 1 – WannaCry Ransomware Attack

WannaCry was a major ransomware outbreak in 2017.

It spread rapidly across vulnerable Windows systems.

Working

Vulnerable Computer



WannaCry infection



Files/System affected



Ransom demand



Attempt to spread to other systems

Impact

The attack affected organizations in multiple countries.

One highly publicized impact was disruption to parts of the UK's National Health Service (NHS).

Lessons

Security updates are important.

Organizations need backups.

Network segmentation can reduce spread.

Cybersecurity requires continuous monitoring.

Old/unpatched systems create major risks.

7. Global Struggle with Cyber Security

Introduction

As cyber attacks become more sophisticated, countries, organizations and international bodies face the challenge of protecting digital infrastructure.

The global struggle with cybersecurity refers to the continuous effort to protect information systems against cyber threats while maintaining economic activity, privacy and international cooperation.

Major Challenges

1. Rapidly changing technology

Technology develops faster than many security policies.

2. Sophisticated attacks

Attackers increasingly use:

Automation

Social engineering

Malware

Ransomware

AI-assisted techniques

3. Lack of cybersecurity awareness

Human mistakes remain an important source of security incidents.

4. Critical infrastructure risks

Attacks against:

Electricity

Hospitals

Transportation

Communication

can have large-scale consequences.

5. International jurisdiction

Attackers can operate across borders.

6. Shortage of skilled professionals

Organizations need trained cybersecurity professionals.

Measures Used Globally

Governments

Cybersecurity policies

Cybersecurity agencies

National incident response teams

Cybercrime laws

Organizations

Firewalls

Encryption

Multi-factor authentication

Intrusion detection

Backups

Security monitoring

International cooperation

Information sharing

Joint investigations

Cybersecurity agreements

Capacity building

8. Case Study – Global Cybersecurity Struggle

SolarWinds Supply-Chain Incident

The SolarWinds incident is an important example of the difficulty of defending interconnected digital systems.

Attackers compromised the software supply chain associated with SolarWinds' Orion platform and inserted malicious code into legitimate software updates.

Organizations that installed affected updates could consequently become compromised.

Why it was significant

It demonstrated that:

Organizations can be attacked through trusted suppliers.

Software supply chains are important security risks.

Cybersecurity cannot focus only on an organization's internal network.

International cooperation and information sharing are important.

Lesson

Modern cybersecurity requires protection of the entire ecosystem:

Organization



Software



Supplier



Cloud services



Third-party providers



Users

Security at only one point is not sufficient.

9. International Cooperation to Combat Cyber Crime

Introduction

Because cyber crime can cross national boundaries, a single country cannot always investigate and prosecute a cyber criminal alone.

Therefore, international cooperation is essential.

Major Forms of Cooperation

1. Information Sharing

Countries and cybersecurity organizations exchange information about:

Threats

Malware

Attack techniques

Indicators of compromise

2. Mutual Legal Assistance

A country may request legal assistance from another country to obtain:

Digital evidence

Records

Witness information

Other investigative assistance

3. Joint Investigations

Law enforcement agencies from different countries may cooperate on international cybercrime investigations.

4. Extradition

If legally permitted, a suspected criminal may be transferred to another country for prosecution.

5. International Treaties

International legal frameworks help countries cooperate in cybercrime investigations.

An important example is the Budapest Convention on Cybercrime.

Role of International Organizations

Organizations such as:

INTERPOL

United Nations

Regional law-enforcement organizations

National cybersecurity agencies

contribute to international cybersecurity cooperation.

Benefits

Faster information sharing

Better investigations

Easier evidence collection

Improved prosecution

Better understanding of international threats

Stronger cybersecurity capabilities

Challenges

Different laws

Privacy requirements

Sovereignty issues

Different legal procedures

Language barriers

Evidence-sharing restrictions

Political and diplomatic considerations

10. Case Study – International Cooperation Against Cyber Crime

Operation Against Emotet

Emotet was a major malware/botnet operation that affected systems internationally.

Law enforcement agencies from several countries cooperated to disrupt its infrastructure.

General process

Cyber criminals



Emotet infrastructure



Victims in multiple countries



International investigation



Law-enforcement cooperation



Infrastructure disruption

Importance

This demonstrates that international cybercrime investigations can involve:

Multiple countries

Technical investigation

Evidence sharing

Coordinated law enforcement

Infrastructure disruption

Lesson

Cyber criminals may operate globally, so law enforcement must also cooperate globally.

11. Planning of Cyber Offences by Cyber Criminals

This is another very important 10–15 mark question.

Definition

Planning of a cyber offence refers to the process through which criminals prepare, organize and execute a cyber attack.

Cyber criminals generally do not randomly attack every target. Many attacks involve some level of preparation.

Stages of Planning a Cyber Offence

Stage 1 – Target Selection

The criminal first identifies a target.

Possible targets:

Individual

Company

Government organization

Bank

Website

Network

Factors criminals may consider

Value of information

Security weaknesses

Financial potential

Accessibility

Stage 2 – Information Gathering

The attacker collects information about the target.

This is often called reconnaissance.

Information may include:

Publicly available information

Organization structure

Technology used

Public contact information

Domain information

Example

An attacker may examine publicly available information about an organization's employees and technology.

Stage 3 – Identifying Vulnerabilities

The attacker looks for weaknesses in:

Software

Systems

Network configurations

Human behavior

Authentication practices

The objective is to identify a possible entry point.

Stage 4 – Choosing Attack Method

Depending on the target, criminals may choose methods such as:

Phishing

Malware

Credential theft

Social engineering

Exploitation of vulnerabilities

Account takeover

Stage 5 – Preparation

The attacker prepares the required resources.

These may include:

Malicious software

Fake communication

Compromised accounts

Infrastructure used to conduct the attack

Stage 6 – Initial Access

The attacker attempts to gain unauthorized access.

Common examples:

Phishing



Victim provides credentials



Attacker obtains account access

or

Unpatched system



Vulnerability exploited



Unauthorized access

Stage 7 – Establishing Persistence

After obtaining access, an attacker may attempt to maintain unauthorized access.

The goal is to prevent losing access if the initial entry point is closed.

Stage 8 – Privilege Escalation

The attacker may attempt to obtain higher privileges.

For example:

Normal account



Higher privileges



Administrative access

Higher privileges can provide greater control over systems.

Stage 9 – Achieving the Objective

The attacker's objective depends on the type of crime.

Examples:

Financial crime

Steal money.

Data theft

Steal confidential information.

Ransomware

Disrupt access to data and demand payment.

Espionage

Obtain sensitive information.

Sabotage

Disrupt systems or services.

Stage 10 – Concealing Activity

Criminals may attempt to make investigation difficult by:

Using compromised infrastructure

Deleting or manipulating evidence

Using multiple systems

Hiding malicious activity among legitimate traffic

For exam purposes, remember this as the concealment stage.

Easy Flowchart for Exam

Draw this:

Target Selection



Information Gathering



Vulnerability Identification



Attack Method Selection



Preparation



Initial Access



Persistence



Privilege Escalation



Attack Objective



Concealment

This flowchart can significantly improve a 10–15 mark answer.

Complete Example

Consider a hypothetical phishing-based attack against an organization.

Step 1

Criminal selects an organization.

Step 2

Public information about employees and the organization is collected.

Step 3

The criminal identifies weaknesses in security awareness or authentication.

Step 4

A deceptive communication is prepared.

Step 5

The message is sent to the target.

Step 6

If credentials are compromised, unauthorized account access may occur.

Step 7

The criminal may attempt to maintain access.

Step 8

The criminal may seek additional privileges.

Step 9

Data or money may be targeted.

Step 10

The criminal may attempt to conceal activity.

Topic Main Points to Remember

Cyber Crime Definition Crime using/against computers or networks

Origins Computers → networks → Internet → modern cyber crime

Against Individual Identity theft, phishing, stalking, bullying

Against Property Data theft, ransomware, IP theft, website attacks

Against Government Espionage, infrastructure attacks, government system attacks

Against Society Large-scale fraud, illegal content, disruption

Global Perspective Cyber crime is worldwide

Transnational Crime Attacker, server and victim can be in different countries

Global Cybersecurity Countries continuously defend digital infrastructure

International Cooperation Information sharing, treaties, joint investigation, legal assistance

Planning Cyber Offence Target → Reconnaissance → Vulnerability → Access → Objective
→ Concealment

★ How to Write a 10–15 Mark Answer

For almost every topic in this unit, use this structure:

1. Definition – 2 marks

Start with a clear definition.

2. Introduction – 1–2 marks

Explain why the topic is important.

3. Main explanation – 5–7 marks

Use headings + numbered points.

4. Example/Case Study – 2–3 marks

Give a relevant real-world example such as WannaCry, SolarWinds or Emotet where appropriate.

5. Diagram/Flowchart – 1–2 marks

Draw a simple flowchart.

6. Conclusion – 1 mark

End with 2–3 strong lines.

Golden formula:

Definition → Introduction → Types/Stages → Explanation → Example → Effects → Diagram → Conclusion

This format is particularly useful for your GTU-style 10–15 mark questions because you can expand or shorten each section according to the marks.

If you are preparing this specifically for your mid-sem, the highest-value topics to learn thoroughly from this unit are classification of cyber crime, transnational nature of cyber crime, international cooperation, global cybersecurity struggle, and planning of cyber offences.

